Project Charter: HEOR Flagship Synthetic RWE Pipeline

1. Executive Summary & Strategic Framework

The objective of this project is to develop a reproducible synthetic Real-World Evidence (RWE) pipeline focusing on Type 2 Diabetes Mellitus (T2DM). The project is designed as a methods-development complement to retrospective oncology work, providing a privacy-free environment to master study design, cohort logic, and payer-oriented evidence framing.

1.1 Governing Strategy

- Core Scope: One central repository utilizing T2DM as the flagship disease area.
- Primary Goal (Year 1): A faculty-ready poster by June 2027.
- Definition of "Done" (Phase 1):
    - Locked study protocol.
    - Reproducible cohort counts and attrition diagram.
    - Baseline characteristics table and treatment-pattern figures.
    - One survival/utilization figure and a synthetic cost summary.
    - A transparent limitations section regarding the synthetic nature of the data.

1.2 Clinical Rationale: Why T2DM?

T2DM is selected over alternatives (e.g., AFib) due to its strategic alignment with managed care and industry HEOR trajectories. The metabolic disease space—specifically SGLT2 inhibitors, GLP-1 receptor agonists, and their roles in cardiovascular/CKD protection—is central to current pharmacoeconomic and formulary conversations, providing stronger signaling for fellowship applications.

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2. Clinical Study Design (Phase 1)

2.1 Study Objective

A retrospective longitudinal incident-cohort study to characterize the treatment and monitoring patterns of adults with T2DM using synthetic data.

2.2 Cohort Logic & Eligibility

Population: Adults (≥18 years) with a first qualifying T2DM diagnosis following a baseline observation window.

Selection Flow:

1. Raw Synthea Population → Initial N
2. Inclusion 1: Diagnosis of T2DM (SNOMED 44054006)
3. Inclusion 2: Age ≥ 18 at index date
4. Inclusion 3: First qualifying T2DM index after baseline observation window
5. Inclusion 4: ≥ 365 days of observable history before index
6. Inclusion 5: Diabetes-relevant monitoring/lab context (if available)
7. Exclusion 1: Prior Type 1 Diabetes (SNOMED 46635009)
8. Exclusion 2: Pregnancy or gestational diabetes window (SNOMED 77386006)
9. Final Analytical Cohort → Final N

2.3 Vocabulary Mapping

Concept	Vocabulary	Code(s)	Use Case
Type 2 Diabetes Mellitus	SNOMED-CT	44054006	Target Condition
Type 1 Diabetes Mellitus	SNOMED-CT	46635009	Exclusion
Pregnancy	SNOMED-CT	77386006	Exclusion/Control
Heart Failure Hosp.	SNOMED-CT	85232005	Event Component
Myocardial Infarction	SNOMED-CT	57054005	MACE Component
Ischemic Stroke	SNOMED-CT	230690007	MACE Component
Metformin	RxNorm	860974	Baseline Context
SGLT2 Inhibitors	RxNorm	1537213, 1373456	Treatment Tracking
GLP-1 Receptor Agonists	RxNorm	1991302, 1151131	Treatment Tracking

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3. Technical Architecture & Blueprint

3.1 The Technical Stack

- Data Layer: Polars (Primary), Parquet (Analytical Store).
- Query/QA Layer: DuckDB (Ad hoc SQL) and SQLite (OMOP-lite relational layer).
- Analysis: lifelines (Survival), statsmodels (GLM/Counts).
- Interface: Typer CLI (Pipeline execution), Quarto (Parameterized reporting).

3.2 Repository Structure

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heor-synth/
├── data/                 # raw, bronze, silver, gold layers
├── configs/              # study, concepts, scenarios (YAML/JSON)
├── sql/                 # extraction scripts (e.g., extract_t2dm_cohort.sql)
├── src/
│   ├── ingest/           # Raw CSV to Parquet
│   ├── normalize/        # Normalization logic
│   ├── omop_lite/        # SQLite mapping layer
│   ├── cohorts/          # Eligibility & attrition logic
│   ├── features/         # Baseline & temporal assembly
│   ├── analysis/         # Stats, survival, and causal inference
│   ├── economics/        # Markov models & ICER
│   ├── reporting/        # Quarto/Streamlit logic
│   └── cli/              # Typer implementation
├── tests/                # Temporal & mathematical validation
├── reports/              # Generated PDF/HTML outputs
└── notebooks/            # Exploratory analysis
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Phase	Focus	Key Deliverables	Timing
Phase 1	Cohort Engine	OMOP-lite mapping, Attrition table, Baseline table, Survival/Utilization result.	Pre-matriculation → Session 1
Phase 2	Comparative Methods	Propensity Score Matching (PSM), IPTW, SMD checks, Kaplan-Meier curves.	Session 2 / Breaks
Phase 3	Economic Layer	Markov cohort trace, ICER calculations, Budget Impact simulation.	Post-Phase 2
Phase 4	Reporting Interface	Read-only reporting layer (Quarto or Streamlit).	Optional/Final

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4. Methodological Rigor & Validation

4.1 Propensity Score Matching (Phase 2)

To demonstrate methodological competence, the pipeline will implement:

- Logistic-regression propensity estimation.
- 1:1 nearest-neighbor matching with a prespecified caliper.
- Standardized Mean Difference (SMD) calculation: $SMD = \frac{x_1 - x_0}{\sqrt{\frac{s_1^2 + s_0^2}{2}}}$

4.2 Economic Validation (Phase 3)

The Markov cohort trace will be validated using a conservation check to ensure the population is preserved across cycles: assert pytest.approx(cycle_sum, abs=1e-5) == 1000.0

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5. Academic Positioning & Deliverables

5.1 Poster Strategy

The project will be presented as a Methodological Validation, focusing on the pipeline's ability to produce reproducible RWE.

Abstract Component Matrix:

Section	Strategy
Background	RWD access is a barrier; synthetic data provide a privacy-free sandbox for algorithm development.
Objective	Engineer an open-source, OMOP-lite T2DM pipeline for reproducible cohort and economic modeling.
Methods	Map Synthea → Parquet → SQLite; apply deterministic inclusion/exclusion criteria; generate cost and survival outputs.
Results	Report cohort counts, attrition diagrams, baseline balance, and synthetic-cost context.
Conclusions	Synthetic OMOP environments are effective for validating HEOR architectures before IRB-gated data access.

5.2 Professional Framing

- To Pharmacy Faculty: "Building a reproducible synthetic RWE pipeline to master cohort construction and payer-oriented workflows in a no-PHI environment."
- To Research Mentors: "A controlled environment to pre-specify cohort logic and analytic workflows, complementing my retrospective oncology work."
- To Fellowship Committees: "Developed a payer-relevant evidence-generation portfolio from the ground up, integrating metabolic disease analytics and economic modeling."

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6. Risk Mitigation

- Avoid "Performative Sophistication": Do not implement causal inference or dashboards until the deterministic cohort engine is stable.
- Avoid Economic Overreach: Budget impact models must only be built *after* resource-use layers are defensible.
- Avoid Architecture Bloat: Stick to "OMOP-lite" (lightweight relational mapping) rather than a full-scale institutional CDM implementation.